OPERATING SYSTEMS LAB

Practical Quiz — Linux Terminal & File Management

Student Name: Uzma Mumtaz	Roll No.: _I-SE9244426
Duration: 40 minutes	Total Marks: 20

Instructions

- Perform all tasks on a Linux terminal. Use your own roll number in the main folder name.
- Write the commands you used in the spaces provided and show the final output to the instructor.
- Do not use a graphical file manager. Each task builds on the previous task.
- Work inside your home directory; do not use sudo or change files outside your quiz directory.

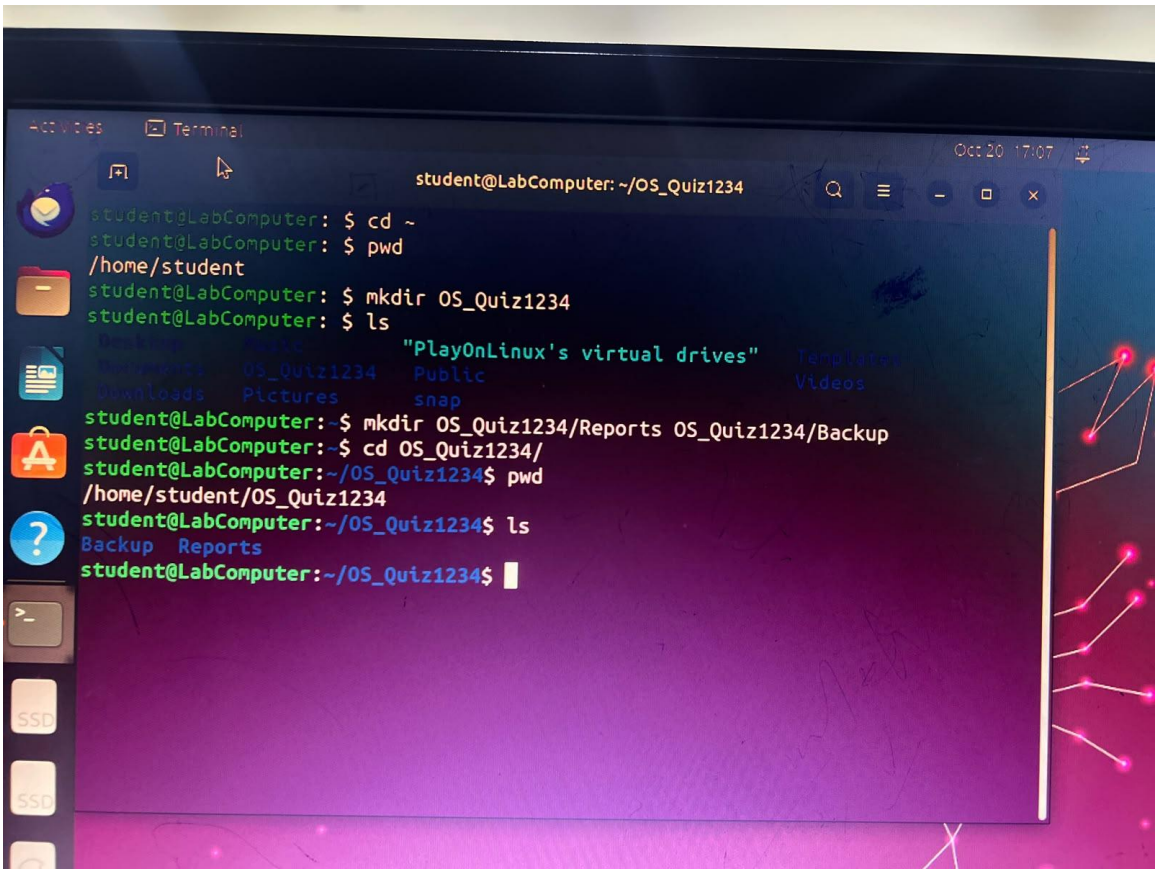
Scenario

You are helping organize the files for a university Operating Systems lab. Create and maintain a small directory structure, record student data, find a specific entry, and control access to a report using Linux commands.

Task 1 — Create the Workspace (4 marks)

Create a directory named OS_Quiz_<RollNo> in your home directory. Inside it, create two directories named Reports and Backup. Navigate into the main quiz directory.

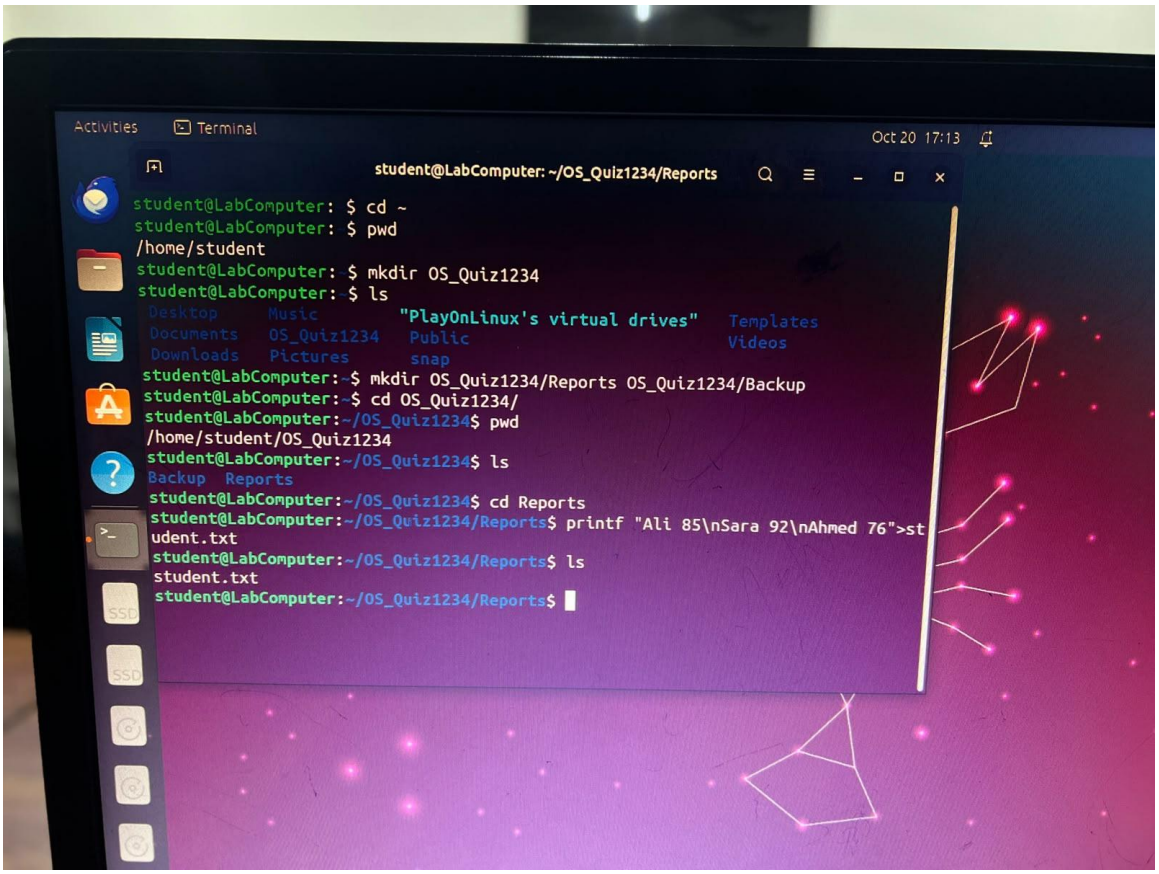
Commands / observations:



Task 2 — Create and Inspect a File (4 marks)

Inside Reports, create a file named students.txt containing the following three lines exactly:
Ali 85
Sara 92
Ahmed 76
Display the file contents in the terminal.

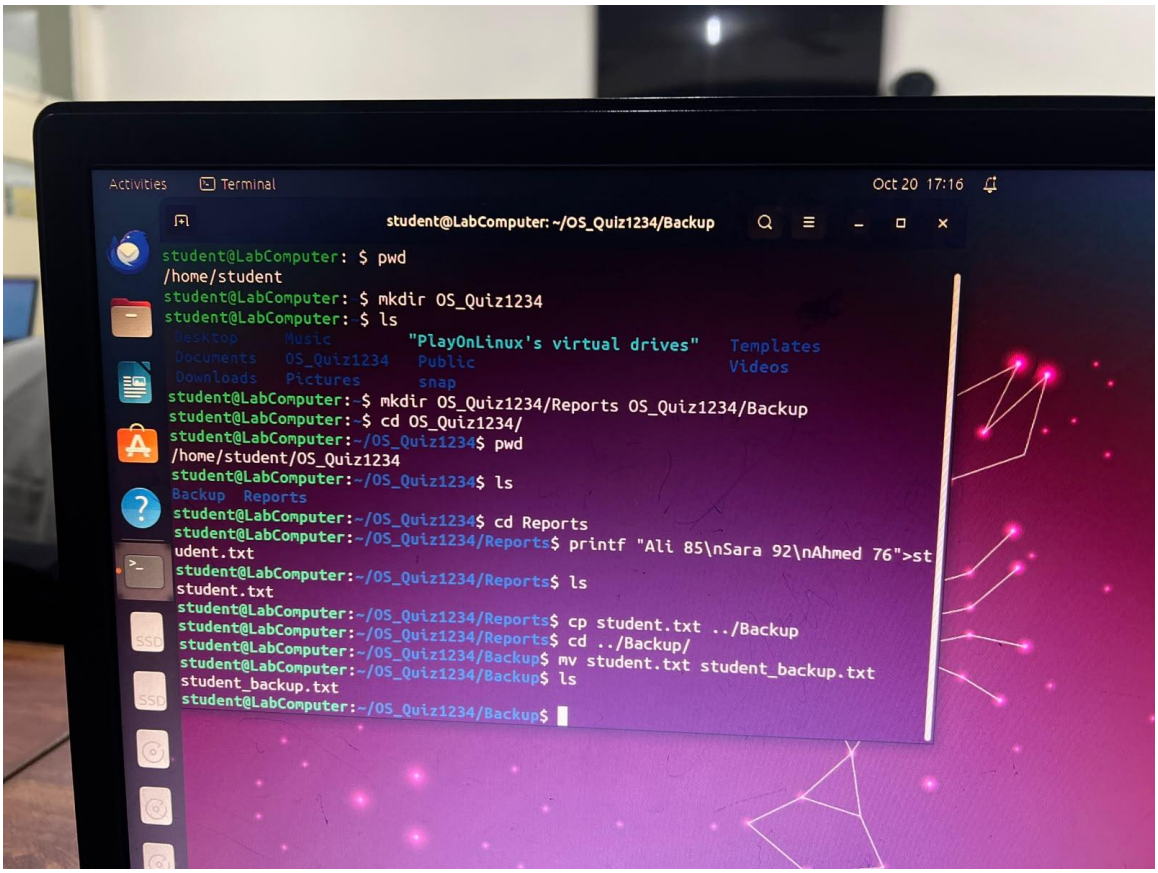
Commands / observations:



Task 3 — Copy and Rename (4 marks)

Copy students.txt from Reports to Backup. Rename the copied file in Backup to students_backup.txt. Show the files in Backup using a terminal command.

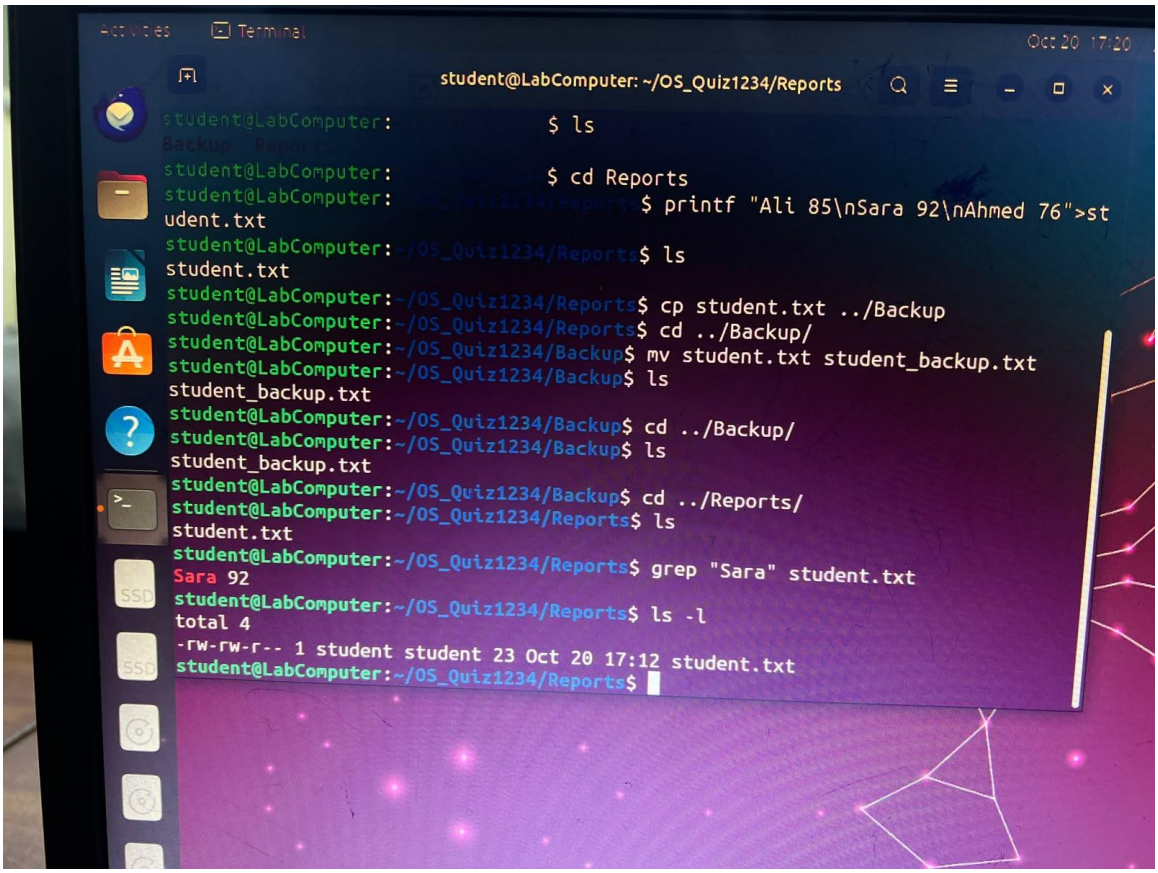
Commands / observations:



Task 4 — Search and Filter (4 marks)

Using a Linux text-search command, display only the line containing Sara from Reports/students.txt. Then display a detailed listing of the Reports directory.

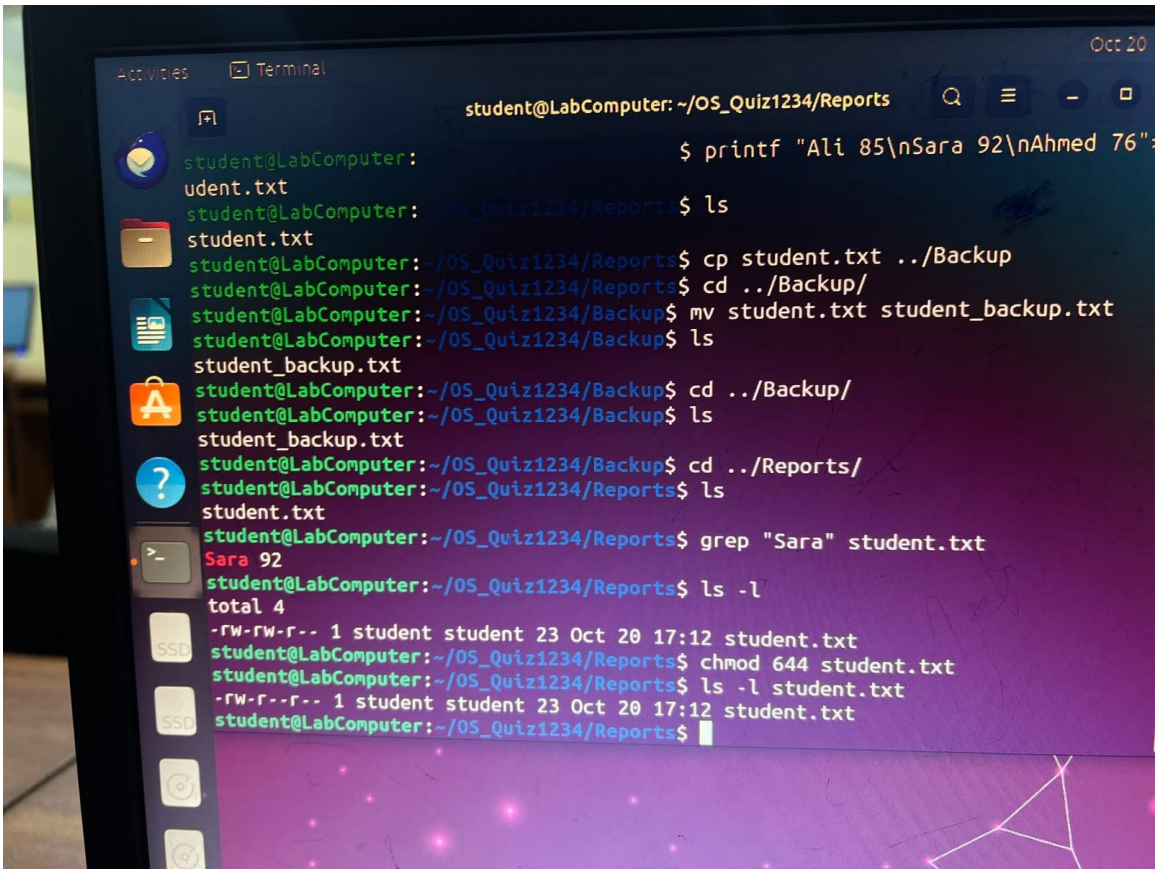
Commands / observations:



Task 5 — Set Permissions and Verify (4 marks)

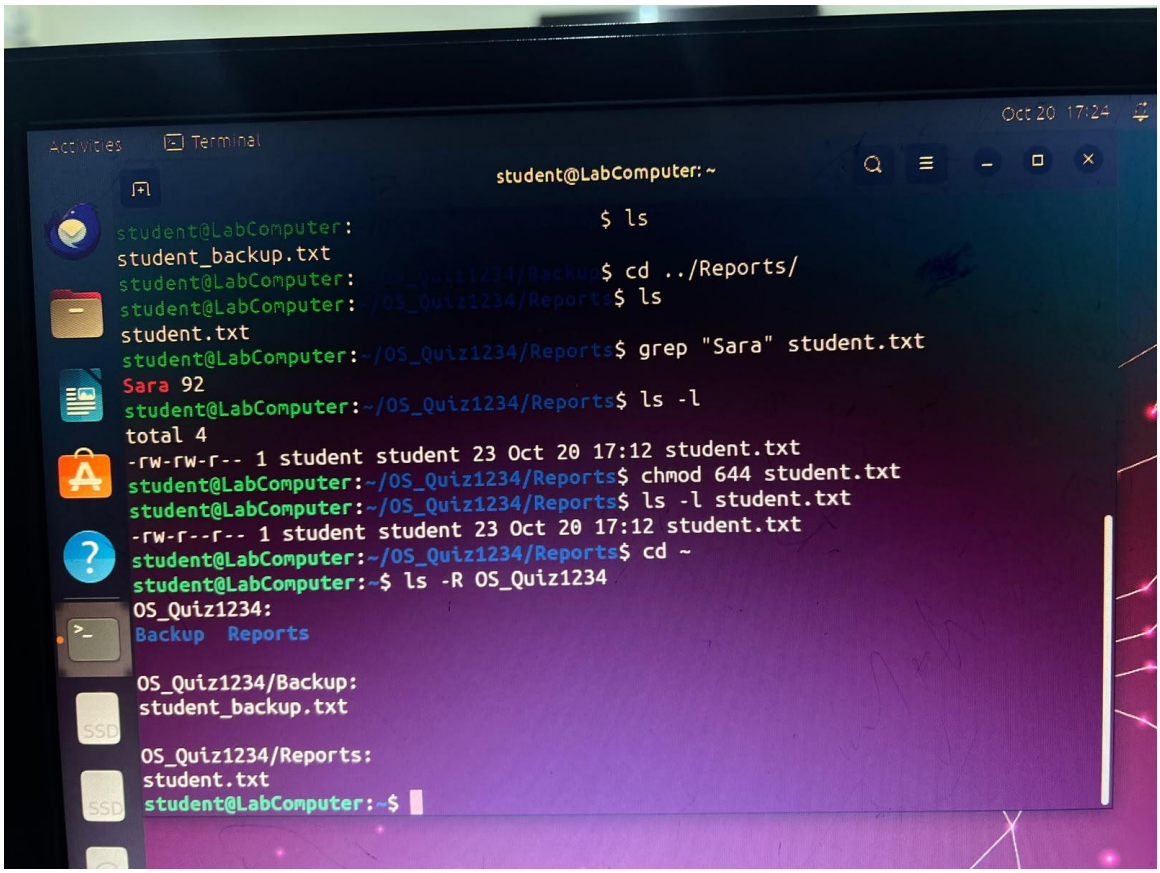
Change the permissions of Reports/students.txt so that its owner can read and write it, while group and others can only read it. Verify the resulting permissions using a Linux command.

Commands / observations:



Submission Checklist

Show your final directory tree (or equivalent recursive listing), the search result, and file permissions to the instructor.


Activities
Terminal
Oct 20 17:24
student@LabComputer: ~
$ ls
student_backup.txt
$ cd ../Reports/
$ ls
student.txt
$ grep "Sara" student.txt
Sara 92
$ ls -l
total 4
-rw-rw-r-- 1 student student 23 Oct 20 17:12 student.txt
$ chmod 644 student.txt
$ ls -l student.txt
-rw-r--r-- 1 student student 23 Oct 20 17:12 student.txt
$ cd ~
$ ls -R OS_Quiz1234
OS_Quiz1234:
Backup Reports
OS_Quiz1234/Backup:
student_backup.txt
OS_Quiz1234/Reports:
student.txt